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HOUSE OF ASPIRANTS

COMPUTER

CHAPTER 9A

Primary Memory

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EDITION
October 2026

LANGUAGE
Punjabi & English

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CHAPTER 9A

Primary Memory

Computer Awareness

IN THIS CHAPTER

- RAM and volatility
- SRAM and DRAM
- SDRAM and DDR
- ROM and firmware
- PROM, EPROM and EEPROM
- Flash memory

- Program Counter stores the next instruction address.
- Accumulator stores processing results.
- Address Bus is unidirectional.
- Data Bus is bidirectional.

MOST IMPORTANT FACTS FOR COMPETITIVE EXAMS

1.Fastest Memory → Register

2.Second Fastest Memory → Cache

3.Smallest Memory Unit → Bit

4.1 Byte = 8 Bits

5.ASCII → 7 Bits

6.EBCDIC → 8 Bits

7.Program Counter stores next instruction address

8.Accumulator stores results

9.Cache stores frequently used data

10.Flip-Flop stores 1 bit

PRIMARY MEMORY (MAIN MEMORY)

Primary Memory is the memory directly accessible by the CPU. It is also known as:

- Main Memory
- Internal Memory
- Semiconductor Memory

- Primary memory stores data and instructions that are currently being used by the CPU. Primary Memory
- RAM (Random Access Memory)
- ROM (Read Only Memory)

RAM (RANDOM ACCESS MEMORY)

It is the working memory of a computer where data, instructions, and program results are temporarily stored while the computer is running.

RAM is usually installed on the motherboard and is directly accessed by the CPU.

Why is it called Random Access Memory?

Because any memory location can be accessed directly without following a sequence.

Example:

The CPU can directly access the 500th memory location without reading the previous 499 locations.

Features of RAM

- Read and Write Memory
- Volatile Memory
- Temporary Storage
- Fast Access Speed
- Directly Accessible by CPU
- Located on Motherboard

What is Volatile Memory?

Volatile memory loses its contents when computer power is switched off. Example

If you are typing a document and suddenly electricity goes off before saving, unsaved data stored in RAM is lost.

Working of RAM

When a computer is switched on:

1.Operating System is loaded from storage (HDD/SSD) into RAM.

2.Application programs are loaded into RAM.

3.CPU processes data directly from RAM.

When the computer is switched off, RAM loses all stored information.

Importance of RAM

The performance of a computer largely depends upon RAM capacity.

Types of RAM

- RAM
- SRAM
- DRAM

1.STATIC RAM (SRAM)

SRAM stands for Static Random Access Memory.

It stores data using flip-flops and does not require continuous refreshing.

Characteristics of SRAM

- Faster than DRAM
- More expensive
- Power consumption depends on capacity, technology and workload; SRAM is not universally lower-power than DRAM.
- Does not require refreshing
- Used in Cache Memory

Uses

- Cache Memory
- High-speed memory applications

Important Exam Point

Cache Memory is generally made of SRAM.

2.DYNAMIC RAM (DRAM)

DRAM stands for Dynamic Random Access Memory.

It stores data using capacitors.

Since capacitors gradually lose charge, DRAM must be refreshed continuously.

Characteristics of DRAM

- Slower than SRAM
- Less expensive
- Higher storage capacity
- Requires periodic refreshing
- Used as Main Memory

Uses

- Main Memory (RAM modules)
- Personal Computers
- Laptops

Important Exam Point

Main Memory is generally made of DRAM. TYPES OF DRAM

1. Asynchronous DRAM (ADRAM)

- Older type of DRAM
- Does not synchronize with CPU clock
- Slower performance

2. Synchronous DRAM (SDRAM)

- Synchronized with CPU clock
- Faster than Asynchronous DRAM
- Widely used in older computer systems

3. DDR SDRAM

DDR = Double Data Rate

It transfers data on both rising and falling edges of the clock signal.

Advantages

- Faster data transfer
- Better performance
- Lower power consumption in modern versions

Types

- DDR
- DDR2
- DDR3
- DDR4
- DDR5

Speed Order DDR5 > DDR4 > DDR3 > DDR2 > DDR

SRAM VS DRAM

Feature	SRAM	DRAM
Full Form	Static RAM	Dynamic RAM
Speed	Faster	Slower
Cost	Expensive	Cheaper
Refreshing	Not Required	Required
Storage Capacity	Lower	Higher
Power Consumption	Depends on implementation	Depends on implementation
Uses	Cache Memory	Main Memory

REGISTER VS CACHE VS RAM

Feature	Register	Cache	RAM
Location	Inside CPU	Between CPU & RAM	Motherboard
Speed	Fastest	Very Fast	Fast
Size	Smallest	Small	Large
Cost	Highest	High	Moderate
Access Time	Lowest	Low	Higher

ROM (READ ONLY MEMORY)

ROM is non-volatile memory that retains startup and firmware instructions without power. Reprogrammable ROM technologies can be updated.

Why is it called Read Only Memory?

Traditionally, data stored in ROM could only be read and not modified by users.However, modern ROM technologies allow limited rewriting.

Features of ROM

- Non-Volatile Memory
- Permanent Storage
- Stores Boot Instructions
- Stores Firmware
- Retains Data Without Power

What is Non-Volatile Memory?

A memory that retains data even after power is switched off.

Example

Mobile phone settings remain saved even after turning off the phone.

BOOTING PROCESS

When a computer is switched on ROM provides startup instructions.

- BIOS is executed.
- Hardware is checked.
- Operating System is loaded into RAM.
- Computer becomes ready for use.

Important Exam Point

ROM contains instructions required during booting.

BIOS (Basic Input Output System)

BIOS firmware is stored in non-volatile memory, commonly flash ROM in modern PCs.

Functions of BIOS

- Starts the computer
- Checks hardware components
- Loads operating system
- Controls basic input/output operations

FIRMWARE

Firmware is software embedded in a device and stored in non-volatile memory; many devices allow firmware updates.

It helps hardware communicate with the operating system and other devices.

Examples:

- Keyboard Firmware
- Hard Disk Firmware
- Printer Firmware
- Router Firmware

TYPES OF ROM

- ROM
- MROM
- PROM
- EPROM
- EEPROM
- Flash ROM

1. MROM (MASKED ROM)

Characteristics

- Oldest ROM technology
- Programmed during manufacturing
- Cannot be modified

Status

Mask ROM is programmed during manufacture and remains useful in specialized high-volume products.

2. PROM (PROGRAMMABLE ROM)

PROM stands for Programmable Read Only Memory

Characteristics:

- Manufactured blank
- User programs it once
- Cannot be modified later

Other Name :One-Time Programmable (OTP) Memory

Uses

- Mobile phones
- Medical devices
- RFID tags
- Gaming consoles

3. EPROM (ERASABLE PROGRAMMABLE ROM)

EPROM stands for: Erasable Programmable Read Only Memory

Characteristics

- Can be erased
- Can be reprogrammed
- Erased using Ultraviolet (UV) light

4. EEPROM (ELECTRICALLY ERASABLE PROGRAMMABLE ROM)

EEPROM stands for: Electrically Erasable Programmable Read Only Memory

Characteristics

- Electrically erasable
- Reprogrammable
- No ultraviolet light required
- Can be rewritten thousands of times

Uses

- BIOS Storage
- Embedded Systems

Memory Trick

EEPROM = Erase by Electricity

5. FLASH ROM

Flash ROM is an advanced version of EEPROM.

Characteristics

- Faster than EEPROM
- Erases data in blocks
- High storage capacity
- Most commonly used ROM today

Advantages

EEPROM: 1 Byte erased at a time

Flash ROM: Block of data erased at a time

Therefore Flash ROM is much faster.

Uses

- Smartphones
- SSDs
- Pen Drives
- Memory Cards
- Digital Cameras

KEY DIFFERENCES RAM VS ROM

Feature	RAM	ROM
Full Form	Random Access Memory	Read Only Memory
Nature	Volatile	Non-Volatile
Storage	Temporary	Permanent
Read/Write	Read and Write	Mostly Read
Speed	Faster	Slower
Data Retention	Lost after power off	Retained after power off
Use	Running Programs	Booting & Firmware

QUICK REVISION NOTES

- RAM is volatile memory.
- ROM is non-volatile memory.
- SRAM is faster than DRAM.
- DRAM requires refreshing.
- Cache Memory uses SRAM.
- Main Memory uses DRAM.
- BIOS is stored in ROM.
- Firmware is embedded software.
- PROM can be programmed only once.
- EPROM is erased using UV light.
- EEPROM is erased electrically.
- Flash ROM is faster than EEPROM.

MOST IMPORTANT FACTS FOR COMPETITIVE EXAMS

1.RAM = Volatile Memory

2.ROM = Non-Volatile Memory

3.SRAM is used in Cache Memory

4.DRAM is used in Main Memory

5.BIOS is stored in ROM

6.PROM = One Time Programmable

7.EPROM uses Ultraviolet Light

8.EEPROM uses Electricity

9.Flash ROM is an advanced version of EEPROM

10.DDR5 is faster than DDR4

11.RAM = Runs And Melts (Data disappears after power off)

12.ROM = Remains On Memory

13.PROM = Program Once

14.EPROM = Erase by Light

15.EEPROM = Erase by Electricity

16.Flash = Fast Block Erasing

17.SRAM = Static = Stable = No Refresh

18.DRAM = Dynamic = Needs Refresh